



KEALEE

Design, build, deliver

PROJECT
Indian Queen East — LOT 53 & LOT 54 & LOT 55 & LOT 56
ADDRESS
9588 Fort Foote Rd · 9584 Fort Foote Rd · 9580 Fort Foote Rd · 9576 Fort Foote Rd
JURISDICTION
Prince George's County, Maryland
ZONE
RSF-95
SHEET
FP-100 — Floodplain Concept Study — Existing Conditions, Impact and Mitigation
SCALE
1" = 30'
STATUS
PRELIMINARY
SHEET NO.
1 OF 1
COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

PLAN TYPE AND PREPARATION

PLAN TYPE SITE DEVELOPMENT / FINE GRADING
SHEET FP-100 — Floodplain Concept Study — Existing Conditions, Impact and Mitigation
DISCIPLINE Maryland Professional Engineer
DRAWN BY KEALEE SITE-PLAN ENGINE
DESIGNED BY
CHECKED BY

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

ZONE	RSF-95
TAX / PARCEL ID	—
GROSS TRACT AREA	75,178 SF 1.726 AC
LOT 53 AREA	12,389 SF 0.284 AC
LOT 54 AREA	23,371 SF 0.537 AC
LOT 55 AREA	25,733 SF 0.591 AC
LOT 56 AREA	13,690 SF 0.314 AC
NET LOT AREA	75,184 SF 1.726 AC
LOT 53 FRONTAGE	50' MIN 83.07'
LOT 54 FRONTAGE	50' MIN 173.55'
LOT 55 FRONTAGE	50' MIN 131.07'
LOT 56 FRONTAGE	50' MIN 84.20'
SETBACKS	REQUIRED PROVIDED
FRONT YARD	30' 30'
SIDE YARD	8' 8'
REAR YARD	20' 20'
COVERAGE	MAX PROPOSED
LOT COVERAGE	30% 6.4%
BUILDING FOOTPRINT	— 4,784 SF
DATUM	
HORIZONTAL	EPSG:2248 NAD83
VERTICAL	NAVD88 (feet)

SITE ANALYSIS

- Gross tract area 75,178 SF
- Area of dwelling 4,784 SF
- Wooded area NOT ESTABLISHED
- Floodplain area NOT ESTABLISHED
- Net lot area 75,184 SF
- TOTAL AREA DISTURBED 41,662 SF (AT LEAST)
- SWM REQUIRED — Sec. 32-174(a)(3) YES — DISTURBANCE EXCEEDS 5,000 SF

BUILDING DATA

	G	B	FF	SF	HT	STY
9588 Fort Foote Rd	58.89	50.22	59.22	58.22	20'	2
9584 Fort Foote Rd	55.64	46.97	55.97	54.97	20'	2
9580 Fort Foote Rd	55.67	47.00	56.00	55.00	20'	2
9576 Fort Foote Rd	60.68	52.01	61.01	60.01	20'	2

G: garage slab · B: basement · FF: finished floor · SF: subfloor · HT: height · STY: STOREYS · A DASH IS NOT ZERO; the elevation has not been established and must be set from a field run topographic survey before construction.

DRAINAGE AREA COMPUTATIONS

CATCHMENT	AREA SF	IMPERV.	PPE Q	POST Q	INCREASE	WQV CF
LOT 53	12,389	15.1%	0.49	0.69	0.20	182
LOT 54	23,371	9.4%	0.92	1.16	0.24	263
LOT 55	25,733	8.2%	1.01	1.24	0.23	266
LOT 56	13,690	14.2%	0.54	0.75	0.21	203
PROJECT TOTAL	75,184	10.9%	2.96	3.84	0.88	924

Peak flows are summed for catchments; Q in cfs. WQV is the required water quality volume.

SEQUENCE OF CONSTRUCTION

- Pre-construction meeting 1 DAY
- Obtain necessary permits 2 DAYS
- Notify Miss Utility at 811 at least 48 hours prior to any excavation 1 DAY
- Install stabilized construction entrance and perimeter sediment control 2 DAYS
- Rough grade the lot 5 DAYS
- Construct dwelling 6 MONTHS
- Install water, sewer and storm services and the driveway 10 DAYS
- Fine grade and stabilize all disturbed areas 5 DAYS
- Remove sediment control once the inspector gives written permission 1 DAY

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

- CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION, FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
- ALL EXISTING AND PROPOSED UTILITIES SHOWN PER PG&C CODE SEC. 32-106.
- PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
- ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
- BOUNDARY SHOWN IS THE RECORDED PLAT OF SUBDIVISION — INDIAN QUEEN EAST — BLOCK "E" AND PARTS OF BLOCKS D & F PLAT BOOK WWW (S. P. 40) (MSA: 33289, 32253), 12TH ELECTION DISTRICT, PRINCE GEORGE'S COUNTY, MARYLAND, TAX MAP 133, BLOCK F, GRID E4, PLAT DATED 6 JUNE 1987, APPROVED 28 JUNE 1987 BY THE MARYLAND NATIONAL CAPITAL PARK & PLANNING COMMISSION, PRINCE GEORGE'S COUNTY PLANNING BOARD, M-NCPPC RECORD FILE 5197863. SCALE 1" = 100'. SURVEYOR: W. L. WEEKS, REGISTERED LAND SURVEYOR NO. 384. TOTAL SUBDIVISION AREA 20,981.1 ACRES. PLAT LEGEND: FOR PUBLIC SEWER & WATER SYSTEMS ONLY. TRANSCRIED AND CHECKED FOR CLOSURE. THE PLAT IS THE BOUNDARY SURVEY. NO FIELD TOPOGRAPHY IS INCLUDED; EXISTING GRADE IS COUNTY LIDAR CONTOUR MAPPING.
- CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.
- NO DRIVEWAY CULVERT IS REQUIRED; THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION, NOT AN OPEN DITCH SECTION; GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSSED CURB AT THE APRON PER DPWAT STD. 380.03, NOTE 6. ON LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE	DATE
LICENSE NO.	EXPIRATION DATE
SEAL	

REVISIONS

— NONE —

FLOODPLAIN CONCEPT — EXISTING CONDITION AND IMPACT BY LOT

Flood elevation of record EL 57 ft, FFS-770017 (PG County DPW&T, 16 July 1979) — stated IN THE WSSC DATUM. Design elevations are NAVD88. THE TWO ARE NOT TIED. L&L to EL + 2.5 ft (feet above).

LOT	LOT AREA SF	AREA BELOW EL 57	% BELOW	FIN FLOOR	FF vs EL 57	BASEMENT	PAD LIFT REQ
LOT 53	12,389	11,700	95.7%	59.22	+2.22	50.22	none
LOT 54	23,371	—	—	55.97	-1.03	46.97	3.0 ft
LOT 55	25,733	25,450	98.8%	56.00	-1.00	47.00	3.0 ft
LOT 56	13,690	4,975	36.6%	61.01	+4.01	52.01	none

HOW TO READ THE LIMIT LINE

THE BLUE DASH-DOT LINE IS THE STATED FLOOD ELEVATION CUT THROUGH EXISTING GROUND, TICKS ON THE WET SIDE. A LOT LYING WHOLLY BELOW THE ELEVATION CARRIES NO LINE, BECAUSE THERE IS NO CROSSING TO DRAW — LOT 54 IS 100% BELOW AND HAS NONE; READ THE TABLE, NOT THE ABSENCE OF A LINE.

MITIGATION CONCEPT — RECOMMENDED

- THE BENCHMARKS H&A (03.06), H&A (14.37), H&A (06.41) TO NAVD88 AND ESTABLISH THE WSSC OFFSET. EVERY VOLUME BELOW IS SIZED OFF IT. 2. EXTEND THE EXISTING FORT FOOTE ROAD CULVERT WITH 42 IN RCP CL IM, 162 FT WITHIN THE RECORDED 60 FT STORM DRAIN EASEMENT ON THE LOT 54 / LOT 55 PARTY LINE, AND ENLARGE OR REPLACE THE INLET, THE EXISTING 36 IN RUNS AT 104% OF CAPACITY AT Q10 ON 39.1 AC; THE 42 IN CARRIES 104.6 CFS AGAINST A 100-YEAR DEMAND OF 94.9 CFS. THIS IS THE ONLY MEASURE THAT CAN LOWER THE WATER SURFACE RATHER THAN DISPLACE IT. 3. RAISE THE LOT 54 AND LOT 55 PADS 3.0 FT — APPROXIMATELY 1,000 CY. 4. SLAB-ON-GRADE OR WENTED CRAWL SPACE ON LOTS 54 AND 55, NO HABITABLE SPACE BELOW THE FLOOD ELEVATION. 5. COMPENSATORY STORAGE CUT FROM THE REAR OF LOTS 54 AND 55. 6. LEAVE THE REAR OVERFLOW CORRIDOR AT EXISTING GRADE. 7. RESOLVE THE FLOODPLAIN EASEMENT ON LOTS 53 AND 54.

BLANKET FILL WAS CONSIDERED AND IS NOT RECOMMENDED

FILLING THE TRACT TO EL 58 IS 19,212 CY AND TO EL 60 IS 24,062 CY — FOUR TO FIVE TIMES THE ENTIRE PROJECT EARTHWORK. IT WOULD FILL THE OVERFLOW SWALE THAT FFS-770017 RECORDS RUNNING WITH THE STORM DRAIN EASEMENT ACROSS THE REAR OF LOTS 50 TO 53, WHICH IS CONVEYANCE AND NOT MERELY STORAGE, AND WOULD REQUIRE COMPENSATORY STORAGE THERE IS NO ROOM FOR ON A 1.7 AC TRACT. GRADINGS DOES NOT LOWER A FLOOD ELEVATION; IT RAISES THE GROUND AND MOVES THE WATER.

FEMA STATUS

ALL FOUR LOTS ARE ZONE X, AREA OF MINIMAL FLOOD HAZARD, OUTSIDE THE SFHA, WITH NO PUBLISHED BASE FLOOD ELEVATION — FEMA NPHL, QUERIED 2025-09-10 AT EACH ADDRESS POINT AND OVER THE WHOLE TRACT. FIRM PANEL 24033C2020E, EFFECTIVE 2016-09-16. THE EXPOSURE ON THIS SITE IS THE COUNTY'S OWN FLOODPLAIN OF RECORD, NOT FEMA'S.

LIMITS OF THIS ANALYSIS — READ BEFORE RELYING ON THIS SHEET

THIS IS A CONCEPT STUDY. IT IS NOT A FLOODPLAIN DELINEATION, NOT A NO-RISE CERTIFICATION AND NOT A DETERMINATION. EXISTING GRADE IS COUNTY 2 FT LOAD CONTOUR MAPPING, NOT A FIELD-RUN SURVEY. THE EL LINE IS DRAWN AT THE NUMERIC VALUE OF A WSSC DATUM ELEVATION ON A NAVD88 SURFACE AND ITS TRUE POSITION DEPENDS ON A TIE NOT YET MADE — NOTE THAT FORT FOOTE ROAD ITSELF READS EL 54.0 NAVD88 AT LOTS 54 AND 55, BELOW THE STATED ELEVATION, WHICH INDICATES THE OFFSET IS MATERIAL. FFS-770017 CARRIES NO CROSS SECTIONS AND ITS CONTROLLING REVISION FFS 960004 IS NOT IN THE RECORD. HYDRAULICS SHOWN ARE RATIONAL METHOD AND MANNING FOR CONVEYANCE SIZING — NO BACKWATER, NO WATER SURFACE PROFILE, NO HEC-RAS, EXCLUDED UNLESS SEPARATELY AUTHORIZED. NEW TOPOGRAPHIC OR BOUNDARY SURVEY, FEMA CLOMR OR LOMR, FULL FLOODPLAIN STUDY, DETAILED HEC-RAS MODELING, GEOTECHNICAL, OR INFILTRATION TESTING, WETLAND DELINEATION, FINAL SWM DESIGN, FINAL SEDIMENT AND EROSION CONTROL, CONSTRUCTION DOCUMENTS, UTILITY RELOCATION, CONSTRUCTION COST, AGENCY FEES, MORE THAN ONE REVISION CYCLE, AND ANY GUARANTEE THAT THE LOTS WILL BE REMOVED FROM THE FLOODPLAIN.

CONCLUSION

FEASIBLE WITH MODIFICATIONS, SUBJECT TO TWO VERIFICATION GATES — (1) THE DATUM TIE, AND (2) A NO-RISE DEMONSTRATION FOR THE CULVERT UPGRADE AND THE PAD FILL. IF GATE 2 CANNOT BE MET, THE FALLBACK IS TO RAISE THE FLOORS AND PLACE NO FILL BELOW THE FLOOD ELEVATION; THE FLOODPLAIN EASEMENT QUESTION ON LOTS 53 AND 54 IS INDEPENDENT OF THE HYDRAULICS AND MAY GOVERN REGARDLESS.

GRAPHIC SCALE — 1" = 30'

